

New Roads Queries

Time limit: 1.00 s

Memory limit: 512 MB

There are n cities in Byteland but no roads between them. However, each day, a new road will be built. There will be a total of m roads.

Your task is to process q queries of the form: "after how many days can we travel from city a to city b for the first time?"

Input

The first input line has three integers n , m and q : the number of cities, roads and queries. The cities are numbered $1, 2, \dots, n$.

After this, there are m lines that describe the roads in the order they are built. Each line has two integers a and b : there will be a road between cities a and b .

Finally, there are q lines that describe the queries. Each line has two integers a and b : we want to travel from city a to city b .

Output

For each query, print the number of days, or -1 if it is never possible.

Constraints

- $1 \leq n, m, q \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 4 3 1 2 2 3 1 3 2 5 1 3 3 4 3 5	2 -1 4